

[SMARTLOGGER USER GUIDE](#)

SmartLogger Features

A complete overview of everything SmartLogger does — from rapid manual QSO entry to automatic WSJT-X logging, FlexRadio slice tracking, SmartLink support and live radio telemetry, LoTW and eQSL.cc upload and sync, logbook ADIF import, paper QSL card tracking, DXCC entity tracking, space weather, the interactive Grid Map of worked squares, and the Grayline / Spots Globe with real-time 3-D day/night shading and live DX spot visualization.

[Features](#) · [Getting Started](#) · [Entering a QSO](#) · [Callbooks Setup](#) · [LoTW Setup](#) · [eQSL Setup](#) · [QSL Services](#) · [WSJT-X Setup](#) · [Radio Setup](#) · [SmartLink Setup](#) · [DX Cluster Setup](#) · [Rotator Setup](#) · [Maps](#) · [Database Management](#)

QSO entry and logging

The Log a QSO panel provides a keyboard-optimized entry form designed for rapid real-time logging:

- **Required fields** — only Callsign, a valid UTC date/time, and a Band or Frequency are needed to save a contact. Everything else is optional.
- **Callsign validation** — format and prefix checks run on focus loss. Invalid callsigns are outlined in red and the Tab key is blocked until the error is corrected.
- **Frequency auto-fill** — entering a frequency auto-resolves the Band from the IARU band plan for the contact's region. Decimal MHz (`14.250`) and grouped MHz.kHz.Hz (`14.250.000`) are both accepted.
- **Country from prefix** — the contact's entity name (e.g. "United States") is resolved from the callsign prefix and populates the Country field automatically.
- **Live UTC clock** — date and time track the real UTC clock by default. A *Set to now* button snaps the timestamp to the current moment at any point.
- **Keyboard shortcuts** — Tab from Callsign jumps directly to Frequency; Space from any short-text field logs the QSO; Enter from Callsign moves to Frequency.
- **Clear Callsign button** — a dedicated button clears only the callsign field and cancels any pending callbook lookup without wiping the rest of the form.
- **Duplicate detection** — if the callsign already appears in your log, an amber warning banner lists the most recent prior contact. It is informational only — logging is never blocked.
- **Edit / delete** — double-click any row in the log grid to load it into the form for editing. The Delete button asks for confirmation.

Callbook lookups

SmartLogger supports three online callbook providers. When a callsign passes format and prefix validation, SmartLogger queries the configured provider and auto-fills Operator Name, Grid Square, Country, and address fields. QRZ.com additionally returns precise latitude, longitude, and county, which are stored with the QSO and used for more accurate heading and distance calculations.

Provider	Cost	Account	Coverage
HamDB.org	Free	None required	US callsigns, some international
QRZ.com	Paid XML subscription	Yes (free + paid tier)	Worldwide, large biographical database
HamQTH	Free	Yes (free account)	Worldwide, community-maintained

- **Lookup priority** — designate a Primary and Backup provider. If the primary does not respond within 2 seconds, the backup is tried automatically.
- **Smart tab order** — when a callbook lookup fills Operator Name, Country, and Grid Square, the Tab key skips those fields automatically so you can move straight to Frequency.
- **Offline operation** — disable all providers in Settings when operating without internet access. Country (from the built-in prefix table) and Band (from the IARU band plan) still auto-fill locally.
- **Session management** — QRZ and HamQTH use session tokens. SmartLogger establishes and caches the session automatically, re-authenticating transparently when a token expires.
- **Credential security** — QRZ and HamQTH passwords are stored DPAPI-encrypted, unreadable outside your Windows user profile.

See [Callbooks Setup](#) for configuration details.

FlexRadio SDR integration

SmartLogger connects to any FlexRadio FLEX-6000, FLEX-8000 and Aurora radios over the network or a SmartLink connection. The integration is designed for both single-operator and Multiflex (multi-station) setups, and extends to compatible external amplifiers that expose themselves through the open SmartSDR API.

- **Auto-discovery** — SmartLogger listens for FlexRadio broadcast beacons and can discover radios on the local subnet automatically. Direct IP entry and subnet-range scanning are also available.
- **SmartLink WAN connectivity** — SmartLogger integrates with FlexRadio's SmartLink remote access feature, enabling connections to your radio over the internet without a VPN. SmartLogger supports convenient login access to **multiple SmartLink accounts** — each with its own display name, stored credentials, and last-login timestamp. Manage accounts via **Radios → SmartLink Accounts...**: add accounts, connect, set a default, or remove stored credentials. The active account's display name appears in the **Your Station bar** while connected so you always know which account is in use. In the radio picker, a **green dot** indicates a local network radio and a **blue dot** indicates a SmartLink WAN

radio; the *Via* column shows *LAN* or *SmartLink* explicitly. When auto-connect is configured, SmartLogger tries the LAN path first and falls back to SmartLink automatically if the radio is not found locally. Disconnecting from SmartLink retains stored account credentials — reconnect any time without re-authenticating. See [SmartLink Setup](#) for full configuration details.

- **TX slice auto-fill** — when a radio is connected, the Frequency, Mode, and TX Power (W) fields populate from the current TX slice whenever you enter a new callsign. Frequency and Mode are read directly from the slice; TX Power reflects the peak forward power seen during the contact — sourced from the amplifier output when one is connected, or from the radio's own FWDPOWER meter otherwise. A warning banner appears if the selected station has no TX slice defined in SmartSDR.
- **Live slice bar** — all active slices are displayed in a strip below the menu bar, showing letter, frequency, and mode. The TX slice tile is highlighted in red.
- **Multiflex / multi-station support** — in a Multiflex configuration (multiple SmartSDR GUI clients connected to one radio), slice groups are shown per station. A radio button lets you select which station's TX slice drives the frequency auto-fill. This prevents accidentally logging on the wrong station's frequency.
- **Auto-connect on startup** — SmartLogger remembers the last connected radio (by serial number or IP) and reconnects automatically at launch. When SmartLink is enabled, the same auto-connect entry works whether you are on your home LAN or connecting remotely via SmartLink.
- **GUI client monitoring** — if you connect to a radio before SmartSDR is running, SmartLogger enters monitoring mode rather than disconnecting. The moment a SmartSDR client connects to the radio, SmartLogger detects it automatically and brings the slice bar live — no manual reconnect needed. Over SmartLink, live telemetry begins flowing immediately even while waiting for SmartSDR to start.
- **Radio identity display** — the connected radio's nickname, model, and serial number are shown in the slice bar. A second row labeled **Available Resources** lists the radio's current capabilities as they are reported over the connection: the number of active panadapters and slices; **GNSS** or **GPSDO** when the radio's oscillator status reports a precision timing lock; and the amplifier model and serial number when an external amplifier is connected. All entries are populated automatically and update in real time as the radio's status changes.
- **Live radio telemetry** — at connect time SmartLogger queries the radio's metering capabilities and subscribes to the voltage, current, fan speed, temperature, and forward power channels available on the connected radio model. Readings are displayed in a compact flowing panel alongside the radio identity strip. **RF power out** (in watts) is shown first; supply voltages, PA current, fan speed, and PA temperature(s) follow. When the radio reports a precision GPS or GPSDO oscillator, a **GNSS/GPSDO** lock-state indicator is appended at the end of the telemetry panel — see [GPS/GNSS status](#) below for the possible states. Items flow across as many lines as needed to fill the panel width evenly, avoiding empty rows. Temperature readings are shown in both °C and °F. Values update in real time — over LAN; only operator-relevant readings are shown. All readings are cleared automatically on disconnect.

- **External amplifier integration** — SmartLogger automatically detects amplifiers that connect to the radio and expose themselves through the open SmartSDR API. When a compatible amplifier is present, its RF output power is used as the forward power reading in place of the radio's own drive level. This power is displayed as the **RF power out** value in the telemetry panel and is used to populate the **TX PWR** field in the Log a QSO form, ensuring the logged power accurately reflects the total transmitted power at the amplifier output. The amplifier model and serial number are shown in the radio resource strip alongside the radio identity. The only amplifier currently known to integrate with FlexRadio via the SmartSDR open API is the **Power Genius XL**.

GNSS / GPSDO status

When the radio reports a built-in GNSS receiver or GPSDO (GPS-disciplined oscillator), SmartLogger queries the radio's oscillator state every five seconds and displays the current lock status as a **GNSS/GPSDO** entry at the end of the telemetry panel. The indicator also appears in the **Available Resources** row of the radio bar.

State	Meaning
Warming Up	The GPS TCXO is warming up; no satellite lock has been achieved yet.
Free Run	The GPS or GPSDO is not locked to a satellite signal; the oscillator is running freely without GPS discipline.
Pull In	The device has detected satellite signals and is actively synchronizing its internal clock to GPS atomic time.
Locking	The GPS has acquired satellites and is working toward a timing lock.
Locked	The GPS has a good sky view and is fully locked. The radio's reference oscillator is disciplined to GPS.
Holdover	The GPS has lost its sky view and is running in holdover mode — the oscillator is coasting on its last good lock.

The status indicator is shown only when the radio reports a GNSS or GPSDO as present. Radios without this hardware show no GNSS/GPSDO entry in the telemetry panel.

WSJT-X integration

SmartLogger listens for UDP broadcasts from WSJT-X on port 2237 and automatically logs each contact the moment you click Log QSO in WSJT-X.

- **Zero re-keying** — callsign, grid square, frequency, mode, signal reports, name, and timestamp are all received from WSJT-X directly.

- **Grid Square from status messages** — WSJT-X sends the contact's grid square in Type 1 (Status) messages, which may arrive after the initial log event. SmartLogger monitors these follow-up messages and updates the Grid Square field automatically whenever it arrives, so the logged QSO always has the most complete information WSJT-X reported.
- **Band auto-resolution** — the TX frequency (reported in Hz by WSJT-X) is converted to MHz and resolved to a band using the IARU band plan for your operator's region.
- **Station identity** — Station Callsign, My Grid, ITU Zone, and CQ Zone are drawn from the active operator profile in SmartLogger. WSJT-X's own station fields are used as a fallback when no active operator is configured.
- **Unicast and multicast network connectivity** — both IP address types are supported. Leave the multicast group blank for standard same-machine or network IPv4 unicast addressing; enter a multicast group address to share WSJT-X events with multiple applications simultaneously.
- **Resilient listener** — socket errors trigger automatic exponential back-off reconnect. The listener restarts cleanly each time the Settings dialog is closed.

See [WSJT-X Setup](#) for configuration details.

LoTW upload, sync, and logbook import

SmartLogger integrates with ARRL's Logbook of the World (LoTW) in three ways.

- **Upload dialog** — choose *QSL → LoTW → Upload to LoTW...* to open the Upload window. Filter by date range, band, and mode; select individual rows or the entire result set with Ctrl+A; then click the Upload button to hand the batch to `tqs1.exe` for signing and submission. If your TQSL certificate is password-protected, TQSL will prompt for it. The window auto-searches on open (showing all unsent QSOs) and refreshes the grid automatically after a successful upload, showing the new LoTW ↑ sent dates.
- **Sync confirmations** — SmartLogger downloads the LoTW QSL report for your callsign and matches confirmed QSOs back to your local log, stamping each with the LoTW confirmation date.
- **Download from LoTW Logbook (ADIF)** — import a LoTW Status Report ADIF file exported from your LoTW account. New QSOs found in the file are added to your log; QSOs already in your log are matched within ±5 minutes and updated with their LoTW sent and confirmed dates. Choose *QSL → LoTW → Download from LoTW Logbook (ADIF)...*
- **LoTW status column** — the log grid shows a glyph for each QSO: ↑ (uploaded, awaiting confirmation) or a green ✓ (confirmed). Sent and confirmation dates are shown in dedicated columns.
- **Per-operator credentials** — each operator profile stores its own LoTW username, password (DPAPI-encrypted), and station location. Switching operators automatically switches LoTW credentials.
- **Check Credentials** — a test button verifies LoTW credentials before you close Settings.

See [LoTW Setup](#) [eQSL Setup](#) for configuration details.

eQSL.cc upload and sync

SmartLogger integrates with eQSL.cc for electronic QSL card exchange. Upload QSOs directly from a built-in upload dialog, download received confirmations back into your log, and track AG (Authenticity Guaranteed) member status for your contacts.

- **Upload dialog** — choose *QSL → eQSL → Upload to eQSL.cc...* to open the Upload window. Filter by date range, band, and mode; select individual rows or the entire result set; then upload the selection in one click. The window auto-searches on open and refreshes the grid automatically after a successful upload.
- **Sync confirmations** — choose *QSL → eQSL → Sync Confirmations from eQSL...* to download your eQSL inbox and stamp matching local QSOs with their confirmation date. Matching uses Callsign + Date/Time (±30 min) + Band + Mode. The first sync fetches everything; subsequent syncs are incremental via a watermark timestamp.
- **AG Member List** — eQSL's Authenticity Guaranteed programme is a verification tier for award-grade contacts. Choose *QSL → eQSL → Refresh the AG Member List...* to download the current AG member list. An **AG** column in the log grid flags contacts who are AG members.
- **eQSL status columns** — the log grid shows an eQSL ↑ column with the upload date and an eQSL ✓ column (in green) with the confirmation date. Both are exported to ADIF.
- **Per-operator credentials** — each operator profile stores its own eQSL.cc username, DPAPI-encrypted password, QTH nickname, and optional upload-from date (to prevent "Multiple accounts match" errors for QSOs that pre-date the current certificate).
- **Sync watermark** — after each successful sync the watermark is saved so the next run fetches only new confirmations. Clear the watermark in *Settings → eQSL* to force a full re-fetch of the entire inbox.

See [QSL Services](#) for full setup and usage details.

Paper QSL card tracking

SmartLogger tracks physical QSL cards sent and received via post directly in your QSO log — no third-party service required.

- **QSL Sent and QSL Rcvd columns** — the main log grid displays the date a paper card was sent and the date one was received for each QSO. The received date is shown in green.
- **QSL Card Update window** — choose *QSL → Cards* to open a dedicated search and batch-update window. Search your log by callsign, band, mode, QSO date, or card status (all, sent, received, or unsent). Select any number of rows and stamp a sent date, a received date, or both in one operation.

- **Card status filter** — quickly list only QSOs with no card sent yet (*unsent*), only those already sent (*sent*), or only confirmed (*received*) to focus work on a specific part of your card queue.
- **Multi-row selection** — hold Shift or Ctrl to select multiple rows in the results grid, then update all selected QSOs at once.
- **Live refresh** — after each update the results grid automatically re-runs the current search so the updated dates appear immediately without reopening the window.

See [QSL Services](#) for details on the QSL Card Update window.

DX cluster integration

SmartLogger connects to a DX cluster node over telnet, displays incoming DX spots in real time, and can automatically send a spot when you log a new QSO. Spots can also be forwarded directly to the connected FlexRadio for display in the SmartSDR panadapter.

- **Spot Display window** — the *Spots* menu opens a live, scrollable spot table showing UTC, frequency, DX call, band, mode, spotter, and comment. A running counter in the header tracks how many spots have arrived since the window was opened. A **Clear** button in the header clears the list and resets the counter without closing the window. Closing the window resets both the list and the counter.
- **Spotting** — logging a new QSO automatically sends a `DX freq call [comments]` command to the connected cluster. No extra steps required.
- **Self-spot confirmation** — when the cluster echoes your own spot back to you, SmartLogger detects it and shows a confirmation message in the status bar.
- **FlexRadio spot forwarding** — when enabled, received spots are forwarded to the connected FlexRadio via the SmartSDR `spot add` API so they appear as frequency banners in the SmartSDR panadapter. A configurable **Spot Lifetime** (default 600 s / 10 min) controls how long each banner remains on the panadapter; set to 0 for spots that never expire automatically. A 30-second warmup after radio connection prevents stale spots from flooding the display.
- **Click-to-tune** — clicking a row in the Spot Display window instantly tunes the connected FlexRadio's TX slice to the spot's frequency. The **Tune to Spot** button in the Grayline / Spots Globe spot info panel does the same from globe spot markers. When a spot carries no mode, the correct mode is inferred automatically from the IARU sub-band designation for your operator's region. See [Click-to-Tune](#) in the DX Cluster Setup guide for full details.
- **Log form population (opt-in)** — when enabled, click-tuning to a spot also populates the Log a QSO callsign field, triggering callbook lookup, duplicate check, and country inference automatically. Off by default. See [Populate Log a QSO Form on Spot Tune](#) for details.
- **Auto-reconnect** — if the cluster drops the connection, SmartLogger reconnects automatically with exponential back-off (5 s, 10 s ... up to 2 minutes), stopping after 5 consecutive failures.
- **Duplicate suppression** — the same callsign on the same frequency is shown only once per 5-minute window.

- **Band and mode filtering** — filter incoming spots by band (160m through 70cm) and/or mode (CW, SSB, FT8, RTTY, ...). Filters are applied both server-side (`accept/spots` commands sent to the cluster on login) and client-side. Filter settings save automatically the moment any checkbox changes — no Save button is required. The band filter is a strict whitelist: spots whose frequency cannot be resolved to a known band are also rejected when a band filter is active.
- **Multiple nodes** — store up to five cluster node configurations. One is active at a time (shown with a ★). Switching nodes takes effect when Settings is closed.
- **Encrypted passwords** — node passwords are stored DPAPI-encrypted in the settings database.
- **Compatible formats** — works with any node using the standard DX Spider or AR-Cluster spot format. ANSI color codes are stripped automatically.
- **Band auto-resolution** — spot frequencies (in kHz) are resolved to band names using the IARU band plan for your active operator's region.
- **DXTEST verification** — a test button in Settings → Spots sends a `DXTEST` command to the cluster and validates the echo-back to confirm your self-spotting credentials are working correctly.
- **Test spot to radio** — a separate test button sends a test spot directly to the connected FlexRadio and reports the radio's reply, making it easy to verify the forwarding path end-to-end.

See [DX Cluster Setup](#) for full configuration details, including [Click-to-Tune](#) and [Log Form Population](#).

ADIF import and export

- **Export** — saves your entire log (or the active operator's log) to an ADIF 3.1.7 file. Contact fields exported include: CALL, QSO_DATE, TIME_ON, BAND, MODE, SUBMODE, FREQ, RST_SENT, RST_RCVD, NAME, QTH (city/location), GRIDSQUARE, COMMENT, COUNTRY, CNTY, CITY, STATE, ADDRESS, TX_PWR, ITUZ, CQZ, DXCC, LAT, and LON. Station fields include STATION_CALLSIGN, MY_GRIDSQUARE, MY_NAME, MY_CITY, MY_STATE, MY_COUNTRY, MY_RIG, MY_ANTENNA, MY_ITU_ZONE, MY_CQ_ZONE, MY_LAT, and MY_LON. LAT/LON are written when precise coordinates were captured from a QRZ lookup; MY_LAT/MY_LON are written when the active operator profile has latitude and longitude set. LoTW and eQSL confirmation dates (sent and received) are exported when present, preserving your complete QSL status in the file.
- **SSB mode normalization** — SmartSDR reports SSB contacts as `LSB` or `USB` rather than the ADIF-standard `SSB`. On export these are automatically normalized: `LSB` → `MODE=SSB SUBMODE=LSB` and `USB` → `MODE=SSB SUBMODE=USB`. SmartSDR-specific CW modes (`CWL`, `CWU`) export as `MODE=CW`. This normalization is applied automatically and requires no action from the operator.
- **Import** — reads an ADIF file and merges the records into the active operator's log. All standard contact and station fields are imported, including COUNTRY, address details, DXCC entity number, ITU zone, CQ zone, and QSL confirmation dates. LoTW and eQSL dates are imported when their corresponding status flag is `Y` (e.g., `LOTW_QSL_RCVD=Y`). On import, `MODE=SSB SUBMODE=LSB` or `SUBMODE=USB` is stored as `LSB` or `USB` to match SmartSDR's native representation. Duplicate-

detection runs on import. The import is accessible from *File* → *Import ADIF* and the toolbar Import button.

- **DXCC entity field** — the numeric DXCC entity number (1–999) is stored for each QSO. It is imported from and exported to ADIF, and populated automatically when importing a LoTW logbook ADIF file. The value is displayed in the **DXCC** column of the log grid.

Space weather display

The Your Station bar shows live geomagnetic and solar-flux data sourced from the NOAA Space Weather Prediction Center (SWPC):

- **Kp Index** — planetary geomagnetic activity index, fetched from NOAA at startup and updated every 3 hours (at HH:03 UTC). A color-coded dot shows the current storm level: dark green (quiet) through orange and red to purple (G1–G5 storm).
- **Kp trend** — a trend arrow (↑/↓) displayed alongside the Kp Index indicates whether the index rose or fell since the previous reading.
- **A Index** — daily Fredericksburg A-index, a measure of 24-hour average geomagnetic activity. Updated at 00:03 UTC each day.

Antenna rotator integration

SmartLogger connects to rotator controllers over TCP and displays live antenna azimuth in the Your Station bar. Two industry-standard ASCII protocols are supported, covering a wide range of popular hardware.

DCU-1 Rotator Control Protocol

Compatible hardware includes the **microHAM ARCO** and the **Green Heron RT-21**, along with any controller implementing the DCU-1 ASCII command set over TCP.

- **Live Beam Heading** — the current antenna azimuth (e.g. `270`) is displayed in the Your Station bar and updates in real time. The field is hidden when the rotator is disabled.
- **Adaptive polling rate** — SmartLogger polls the controller at 1 Hz when the antenna is stationary, then automatically switches to 5 Hz the moment the heading changes, providing fluid updates while the antenna is repositioning. The rate returns to 1 Hz once the heading is stable for 1 second.
- **Listen-only mode** — some controllers send unsolicited `;xxx` position frames automatically while rotating. Enable *Disable polling* to switch SmartLogger to listen-only: it seeds the initial heading with one `AM1;` query on startup, then listens passively for updates without sending further queries.
- **Automatic mode detection** — if SmartLogger receives an unsolicited position update while in polling mode, it automatically switches to listen-only, checks the *Disable polling* box (shown in red), and saves the setting. No manual configuration required.

- **Connection recovery** — if the controller stops responding, polling halts and a status message is shown. Clicking *Test Connection* in Settings → Rotators restarts the service immediately — no need to close Settings.

Yaesu GS-232A/B Rotator Control Protocol

Compatible with any controller implementing the Yaesu GS-232A/B ASCII command set over TCP, including the **K3NG Rotator Controller** firmware and other compatible hardware.

- **Live Beam Heading** — azimuth is displayed in the same Your Station bar Beam Heading field as the DCU-1 integration. Both protocols contribute to the same field; only one can be active at a time.
- **Polling mode** — the GS-232A/B protocol does not support unsolicited updates, so SmartLogger always polls: it sends a `C;` query each cycle and parses the `AZ=aaa EL=bbb` response. The same adaptive 1 Hz / 5 Hz polling rate used for DCU-1 applies.
- **Test Connection** — verifies the link by sending `C;` and confirming the controller replies with an `AZ=` response. A successful test also restarts the live service immediately.
- **Connection recovery** — same behavior as DCU-1: polling halts on failure and *Test Connection* in Settings → Rotators restarts the service without closing Settings.

See [Rotator Setup](#) for configuration details on both protocols.

QSO heading and distance

SmartLogger computes the great-circle bearing and distance from your station to the contact in real time, using the most precise coordinates available for each end of the path:

- **QSO Heading** — initial bearing in degrees (e.g. `043°`), displayed in the Your Station bar. This is a calculated bearing, independent of the physical antenna position reported by the rotator.
- **Distance** — great-circle distance in km and miles, e.g. `1 234 km / 767 mi`.
- **Contact station coordinates** — precise decimal-degree latitude and longitude returned by a QRZ.com callbook lookup are used when available. The contact's Maidenhead grid square is used as a fallback when no QRZ lat/lon is present.
- **Your station coordinates** — the Latitude and Longitude fields in the active operator profile are used when set. The profile's Grid Square is used as a fallback when lat/lon is not entered in the profile.

Both fields update live as you type. They remain blank until both stations have resolvable coordinates — either a precise lat/lon or a valid 4- or 6-character Maidenhead locator.

Interactive map suite

The **Maps** menu provides six interactive map windows. The five 2D maps show your log's progress in different dimensions — all support pan and zoom, filter by band, and update automatically as you log

QSOs. The Grayline / Spots Globe adds a 3-D view with real-time day/night shading, grayline propagation duct visualization, and live DX cluster spot markers.

- **Grid Map** (*Maps → Grid Map*) — world equirectangular map overlaid with Maidenhead 4-character grid squares, colored **yellow** (worked) or **green** (LoTW confirmed). Includes a **cyan** pending-QSO preview that highlights the grid square as you type it into the Log a QSO form, plus an amber beam-line arrow from your home grid to the pending grid. Requires no internet download.
- **Worked All States** (*Maps → Worked All States*) — map of all 50 US states and DC for tracking the WAS award. The 48 contiguous states appear in the main view; Alaska and Hawaii are shown in separate inset boxes. Includes a Show Labels toggle for 2-letter state codes.
- **DXCC Entity Map** (*Maps → DXCC Entity Map*) — world map with each ADIF DXCC entity colored by worked and confirmed status. Overseas territories are shown as separate regions from their parent country. Tiny island entities that are too small for polygon rendering appear as small colored dots at their geographic center.
- **CQ Zone Map** (*Maps → CQ Zone Map*) — world map of all 40 CQ zones using true geographic zone boundaries, so large countries are correctly split across zone lines. Country borders are drawn over the zone fills as a reference layer. Zone numbers are labeled at each zone's geographic center.
- **ITU Zone Map** (*Maps → ITU Zone Map*) — world map of all 75 ITU zones, also using true geographic boundaries and the same country-overlay reference layer as the CQ Zone Map.
- **Grayline / Spots Globe** (*Maps → Grayline / Spots Globe*) — fully interactive 3-D Earth with real-time astronomically accurate day/night shading, NASA satellite textures, and an asymmetric grayline propagation duct (9° night-side / 4° day-side) centred on the civil twilight terminator. Overlays your worked and confirmed DXCC entities directly on the sphere surface. When the DX cluster is connected, live spot markers appear at DXCC entity locations and expire automatically after the configured spot lifetime. Click any spot marker to open a spot info panel showing callsign, spotter, frequency, band, mode, and time; a **Tune to Spot** button in the panel tunes the connected FlexRadio to the spot with the same click-to-tune intelligence used by the Spot Display window. Click any point on the globe to draw short-path and long-path great-circle arcs with bearing and distance. Requires the Microsoft Edge WebView2 Runtime (included by default on Windows 10/11) and no internet connection at run time.

The five 2D map windows each save their band filter selection, window position, size, zoom level, and pan position independently and restore them the next time the window opens.

See [Maps](#) for full usage details on all six windows.

Operator profiles

SmartLogger supports multiple operator profiles, each with its own QSO database:

- **Per-callsign databases** — each operator's QSOs are stored in a separate SQLite file (`smartlogger-qsos-{CALL}.db`). Switching the active operator instantly swaps the log grid to that operator's history.

- **Station identity** — each profile stores the operator's callsign, grid square, latitude, longitude, IARU region, ITU zone, and CQ zone. These auto-fill the "your station" fields on every new QSO. Entering a precise latitude and longitude gives more accurate heading and distance calculations than the grid square alone.
- **LoTW credentials** — per-operator LoTW username, encrypted password, and station location.
- **Automatic first-operator activation** — the first operator profile you create is set as active automatically.
- **Your Station bar** — the callsign, QTH, and grid square shown in the Your Station bar are populated from the active operator profile and are read-only display fields. To change them, edit the operator profile in *Settings* → *Operators*.

IARU band plans

SmartLogger ships with default IARU Region 1, 2, and 3 band plans and uses them to resolve Band from Frequency automatically and to infer the correct mode when click-tuning to a spot with no mode data. The plans are fully editable:

- **Three regions** — Region 1 (Europe/Africa/Middle East), Region 2 (Americas), Region 3 (Asia/Pacific).
- **Sub-band designations** — each HF band is divided into CW, DIGU (narrow-band digital), and SSB sub-bands following the IARU "Preferred Mode and Usage" designations. When click-tuning to a spot that carries no mode, SmartLogger looks up the frequency against the active operator's regional sub-band table and selects the appropriate mode automatically — CW in the CW portion, DIGU for narrow-band digital, and LSB or USB for the phone portion based on conventional band-plan splits.
- **In-app editing** — add, edit, or delete band plan segments directly in *Settings* → *IARU Band Plans* without touching a configuration file. Rows are color-coded by band so adjacent bands are easy to distinguish at a glance.
- **Per-region reset** — reset any region to the built-in defaults at any time.
- **Narrowest-match wins** — when a frequency falls inside multiple overlapping segments (e.g. a CW sub-band inside a wider all-modes band), the narrowest matching segment is selected.

Database management

SmartLogger stores both QSOs and settings in SQLite databases in `%LOCALAPPDATA%\SmartLogger\`.

- **Integrity check** — runs SQLite's `PRAGMA integrity_check` and displays a detailed result. Green for a clean database, red with details for any problem found.
- **Recover / rebuild** — exports all recoverable rows to a temporary file, deletes the old database, and reimports — equivalent to a full VACUUM into a fresh file. A backup is kept.
- **Open database folder** — opens the SmartLogger data folder in Windows Explorer.

- **Delete all QSOs** — a guarded action in the Danger Zone that wipes the active operator's entire QSO log after explicit confirmation.
- **Standard formats** — SQLite databases and ADIF exports use open, well-documented formats. Your data is never locked in a proprietary format.

See [Database Management](#) for details on each maintenance operation.

SmartLogger user help — see also [Entering a QSO](#) · [Callbooks Setup](#) · [LoTW Setup](#) · [eQSL Setup](#) · [QSL Services](#) · [WSJT-X Setup](#) · [Radio Setup](#) · [SmartLink Setup](#) · [DX Cluster Setup](#) · [Rotator Setup](#) · [Maps](#) · [Database Management](#)

© 2026 BDNI Consulting. All Rights Reserved.